A Complementary Spectral-Spatial Method for Hyperspectral Image Classification

Lulu Shi, Chunchao Li, Teng Li, and Yuanxi Peng

***Abstract*— In hyperspectral image (HSI) classification, using spatial information as a supplement to spectral information is an effective way to improve classification accuracy. In this article, a novel robust complementary method using spectral-spatial information is proposed to reduce the information loss in feature extraction, thus improving the classification effect. In short, two complementary feature extraction stages are used to get the probability maps for decision fusion. In the stage of pre-processing feature extraction, we propose an adaptive cubic total variational smoothing method (ACTVSP), which is first proposed and applied in the remote sensing research field, to obtain the first-stage probability map. At the same time, we utilize edge-preserving filtering in the post-processing stage and obtain the second probability map by means of pixel-level classifier. Finally, the probability-like maps obtained in the above two stages are integrated by decision fusion rules. Experiments on ten public datasets show the effectiveness of our proposed method and demonstrate the superiority of distinguishing different land covers on the basis of very few training samples. Therefore, it can be applied to practical applications in different scenes.**

***Index Terms*— Adaptive cubic total variation, augmented Lagrange multiplier, feature extraction, hyperspectral classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) are made up of hyperspectral imager aboard the aerospace craft to capture 3-D images. Every pixel in the image contains abundant spectral information, space, and radiation, which have characteristic of spectrum close to continuous and mapping unity. These characteristics make HSIs widely used in many fields of actual target classification and recognition. Such as mineral exploration [1], land cover [2], [3], agricultural production [4], [5], and military target detection [6], [7]. At the same time, the dimension of HSI is often much larger than the number of available training samples, which brings many challenges to its classification and recognition applications. HSIs classification is one of the research hotspots in remote sensing.

The goal of classification of HSIs is to assign a category label to each pixel in the image according to the sample characteristics. Since different ground objects have different spectral curves, many methods using spectral features have been proposed for HSI classification. For example, support vector machine (SVM) [8], K-nearest neighbor (KNN) [9], naive Bayes (NB) [10], and sparse representation classification [11] are all based on the spectral features of HSIs. Among them, SVM using kernel transform technology [12] obtains a separate hyperplane through a small number of training samples, and then distinguishes different types of hyperspectral pixels, which avoids the phenomenon of Hughes and improves the classification effect a bit. Unfortunately, the experimental results show that most machine learning classification algorithms based on spectral features are not ideal. The main reason lies in the properties of hyperspectral data. There are "same thing different spectrum" and "foreign matter same spectrum" problems that lead to highly nonlinear, eventually making some classification models based on statistical pattern recognition difficult to deal with the original HSIs.

In order to address the above limitation, a number of researchers have introduced spatial information of HSIs on the basis of spectral information. As in [13] and [14], the spatial correlation between spatial pixels was used as a supplement to spectral information to enhanced and improved the classification performance of per-pixel classifier. The theoretical basis of this operation is that pixels belonging to a small area have a high probability of belonging to the same category, that is, adjacent pixels are associated with a large extent. Besides, many spectral-spatial combined classification methods [15]–[18] have been proposed in the past 10 years, and the experimental results show that when using spectral–spatial information, the accuracy and robustness of classification results are greatly improved. According to the different processing ideas of spatial-spectral information combination, the existing methods can be mainly divided into two classification methods based on pre-processing and post-processing. The classification method based on pre-processing is to extract spatial features through certain structures and rules, then fuse them with spectral features, and finally feed them into the classifier. The spatial feature extraction method based on morphological contour is one of the most representative methods. For example, extended morphological

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profiles (EMPs) [19] have been proposed for constructing spectral-spatial features which are adaptive definitions of the neighborhood of pixels. In [20], invariant attribute profiles (IAPs) have been proposed to identify the same material from different scenes or locations in space. In [21] and [22], a series of open and close operators with different scales are used to extract texture information from images. Furthermore, spectral-spatial kernels were proposed to classify HSIs in the kernel space. In [15], [23], and [24], researchers proposed the use of composite kernel, morphological kernel, and graph kernel to improve SVM classifier [8], respectively.

Differently, the post-processing-based classification method only uses spectral information to pre-classify HSIs by a pixel-by-pixel classifier at first, and then corrects the pre-classification results according to the spatial dependence between pixels to further improve the classification accuracy. For example, in [25], multiple logistic regression (MLR) is first used as a classifier to pre-classify HSIs to obtain a posterior probability, and then a Markov regularization term [26] describing prior probability is used for post-processing to obtain a new classification result map. In [27] and [28], the post-processing is carried out by introducing the total variation regularization term to adaptively adjust the weight of pixels in spatial neighborhood. In [29], based on extended random walk (ERW), the correlation between adjacent pixels is used to optimize the classification probability.

Besides, many spectral-spatial information-based deep methods have been proposed for HSI classification in recent years, such as convolutional neural networks (CNNs) [30]–[32]. In [33], compact and discriminative stacked autoencoder has been proposed to learn a low-dimensional feature space. To address bottlenecks in complex scenarios that require fine-grained categorization, Hong *et al.* [34] provided a baseline solution by developing a general multimodal deep learning (MDL) framework. In [35], a classification method based on multiview deep neural network is proposed to extract the spatial–spectral features of the scene with limited samples. However, compared with traditional machine learning methods, deep learning methods usually require more training samples and more parameter adjustment.

The above two methods are the main ideas of combination of spectral and spatial information. But there is a problem in both pre-processing and post-processing, that is, some information will be lost. The classification method based on pre-processing is to classify the image after feature extraction [36]. In the process of feature extraction, part of the texture information of the image will be lost, especially the target of small size is easy to be ignored. However, the classification method based on post-processing firstly classifies the image at the pixel level, which can avoid the problem of ignoring details of the method based on pre-processing. The accompanying problem is that it will be affected by the noise of the original image and lead to misclassification [37]. Even if the obtained probability map is optimized later, the probability distortion cannot be completely compensated. Therefore, the classification method based on post-processing also loses part of image information. From the above analysis, it can be known that since the information which two methods lost is different, it may be possible to combine them by some ways to reduce the loss.

To overcome the aforementioned issue, the final decision fusion based on the result of pre-processing and post-processing classification method was proposed. Recently, a fusion framework [38] based on pre-processing and post-processing classification method is proposed to improve classification effect. The main motivation of this integration is that the large-scale objects can be extracted in the pre-processing stage, and the small-scale objects can be modeled in the post-processing stage. Although the method uses the spatial correlation of pixels as the basis in the stage of pre-processing feature extraction, the differential processing of spectral information is neglected in the SP method.

Considering the differential processing of spectral information, in this article, an adaptive complementary spectral-spatial method is proposed. To be more specific, our method is mainly divided into two parts. In the first stage, an adaptive cubic total variational smoothing method (ACTVSP) is proposed, which is a pre-processing classification method. It should be pointed out that the adaptive parameters are the embodiment of different treatments for different bands. After the spectral and spatial smoothing of HSIs and feature extraction, a pixel-level classifier SVM [8] is used for classification. The second stage is based on the post-processing classification method. First, the pixel direction classification map obtained by SVM classifier [8] is represented as multiple probability maps (the probability that the pixel belongs to a specific category), and edge-preserving filtering (EPF) [39] is carried out to classify each pixel according to the maximum probability. Finally, the probability-like maps obtained in two stages are integrated by decision fusion rules. The main motivation of the proposed method is that the rough structure profile of HSIs can be extract in the first stage, but details will be ignored and edges contour will be fuzzy. Meanwhile, class probability obtained after edge-preserving filtering optimization in the second stage can enhance the details and edge contour as a supplement to the first stage. To sum up, the main contributions of this article are as follows.

1) An adaptive HSI feature extraction method based on spectral-spatial joint feature is proposed, i.e., ACTVSP. It can smoothen the spatial and spectral directions of the whole HSI while maintaining the geometric characteristics, reducing the impact of noise on classification.
2) A robust complementary spectral-spatial method which absorbs the pre-training thought of deep learning is proposed from a new perspective, which improves the performance of hyperspectral classification effectively. In a word, it is to make up for the information loss of feature extraction with the probability map obtained by post-processing.
3) In order to prove the generalization ability and robustness of the algorithm, experiments are carried out on ten public hyperspectral datasets, including both classical and newer datasets. State-of-the-art classification effects were obtained under a set of fixed parameter values and a fixed number of training samples for all datasets.

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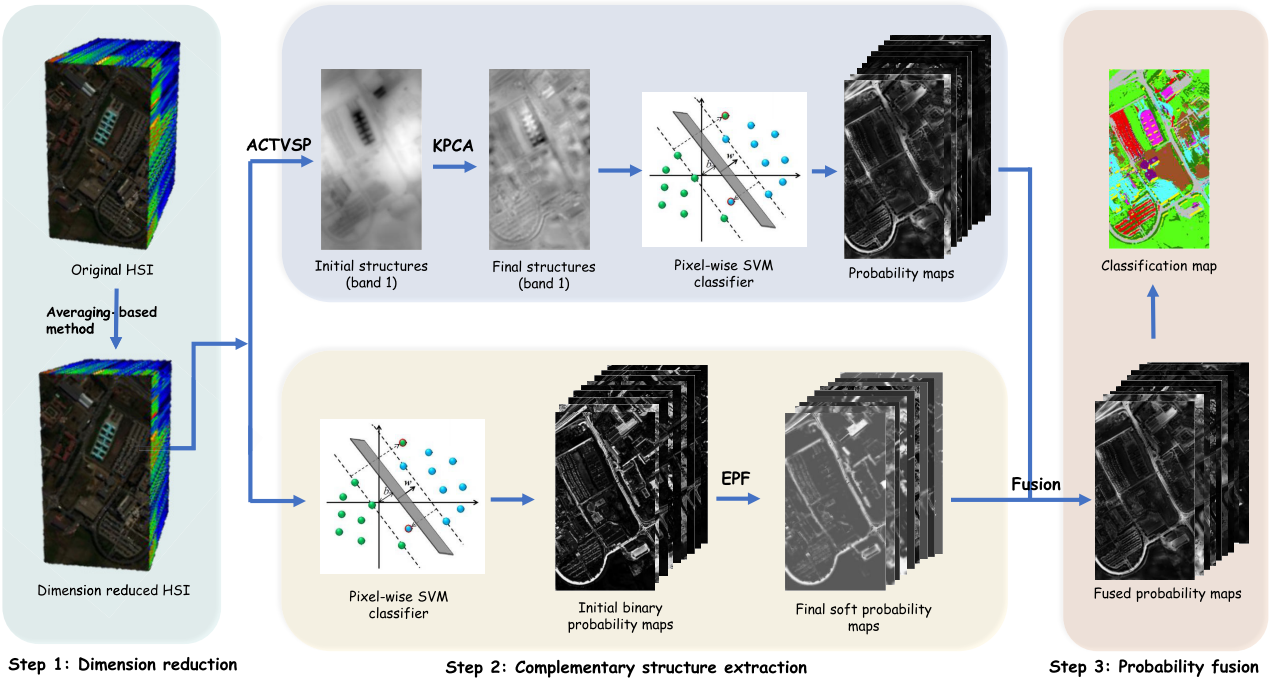

Fig. 1. Schematic of the proposed method.

The rest of this article is organized as follows. Section II introduces the background of the proposed method. Section III introduces the proposed method. Section IV is experiment and analysis. And the conclusion is given in Section V.

II. Background

A. Vectorization Representation

A HSI is a 3-D datacube. In this article, it is expressed as $f(x, y, z)$, where (x, y) and z denote the coordinate in spatial dimension and spectral dimension, respectively.

For the convenience of description, we use matrices and vectors to represent images and variables. For example, suppose that HSI has M rows, N columns, and T bands, we stack the input HSI $f(x, y, z)$ into a column vector of size $MNT \times 1$ according to the lexicographic order. For ease of reading, we use the bold letter $\mathbf{f}$ to represent the vectorized version of the HSI

$$\mathbf{f} = \mathbf{vec}(f(x, y, z)) \tag{1}$$

where **vec** represents the vectorization operator.

B. HSI Modeling

Feature extraction of HSIs faces many challenges. How to model HSIs is an important problem to be considered first. Due to the observed HSIs with a lot of noise [40], this article adopts the following methods to model HSI

$$\mathbf{g} = \mathbf{Hf} + \eta \tag{2}$$

where $\mathbf{f} \in \mathbb{R}^{\mathrm{MNT}\times 1}$ is a vector of size $M \times N \times T$ which denotes the unknown HSI, the matrix $\mathbf{H} \in \mathbb{R}^{\mathrm{MN}\times MN\times T}$ is a 3-D convolution operation for linear transformations, and $\mathbf{g} \in \mathbb{R}^{\mathrm{MNT}\times 1}$ is a vector denoting the observed HSI. $\eta \in \mathbb{R}^{\mathrm{MNT}\times 1}$, is a vector denoting the noise. Our goal is to obtain HSI $\mathbf{f}$ from observed HSI $\mathbf{g}$ using the proposed model by smoothing the HSI.

C. Cubic Total Variation Operator

We define an 3-D total variation operator $\mathbf{D} = [\mathbf{D_x}^T, \mathbf{D_y}^T, \mathbf{D_z}^T]^T$, where $\mathbf{D_x}$, $\mathbf{D_y}$, and $\mathbf{D_z}$ are the first-order forward finite difference operators along the horizontal, vertical, and spectral dimension, respectively. The specific formula is expressed as follows:

$$\mathbf{D_x f} = \mathbf{vec}(f(x+1, y, z) - f(x, y, z)) \tag{3}$$

$$\mathbf{D_y f} = \mathbf{vec}(f(x, y+1, z) - f(x, y, z)) \tag{4}$$

$$\mathbf{D_z f} = \mathbf{vec}(f(x, y, z+1) - f(x, y, z)). \tag{5}$$

In order to control the smoothness of spatial and spectral dimensions more flexibly, this article introduces the following three scaling factors $(\beta_x, \beta_y, \beta_z)$, multiplied by $\mathbf{D_x}$, $\mathbf{D_y}$, and $\mathbf{D_z}$, respectively, and obtains $\mathbf{D} = [\beta_x \mathbf{D_x}^T, \beta_y \mathbf{D_y}^T, \beta_z \mathbf{D_z}^T]^T$. Meanwhile, the total variation norm is defined as

$$\|\mathbf{f}\|_{\mathrm{TV}_2} = \sum_p \sqrt{\beta_x^2 [\mathbf{D}_x \mathbf{f}]_p^2 + \beta_y^2 [\mathbf{D}_y \mathbf{f}]_p^2 + \beta_z^2 [\mathbf{D}_z \mathbf{f}]_p^2} \tag{6}$$

where $[\mathbf{f}]_p$ denotes the pth component of the vector $\mathbf{f}$.

III. Proposed Method

The schematic of this method is shown in Fig. 1, which mainly includes three key steps: dimension reduction, complementary feature extraction, and probability fusion. The details are as follows.

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A. Dimension Reduction

In order to reduce the execution time, we adopt the averaging fusion method to reduce the spectral dimension of HSI for the subsequent steps. Duan *et al.* [41] prove that average dimension reduction is an effective tool in computational efficiency and protection of pixel reflection. Specifically, the original image $\mathbf{g}$ is first divided into B subsets according to spectral dimension. The reduction of the dimension of the data is then performed on each subset.

B. Complementary Spectral-Spatial Adaptive Method

Based on adaptive cubic total variation model [28] for feature extraction can smooth HSI well. However, it is easy to blur the edge profile. It is necessary to choose appropriate parameters to achieve a balance between maintaining the edge contour and smoothing the image. The classification effect obtained by using the ACTVSP method alone for feature extraction is largely dependent on the selection of parameters and tends to blur details and edges. Therefore, the robustness is not strong. The solution of this article is to use the edge-preserving filter [39] to supplement the probability, to compensate for the edge ambiguity in the feature extraction stage, so as to improve the classification accuracy. Besides, the fusion of the two methods will enhance the robustness of the overall method. Although there are parameters that need to be adjusted for edge-preserving filtering, for the two complementary stages, it is only necessary to select the parameters with good overall effect but no need for the best, and the ideal effect can be achieved after combination. Specifically, the method proposed in this article is as follows.

1) Weight Parameter: Assuming the dimension reduced HSI f is a $M \times N \times B$ matrix. It should be pointed out that, for the convenience of explanation, the f mentioned in the description of weight parameters is in matrix form, but the actual operation is to convert f into the corresponding vector $\mathbf{f}$ for processing. Yuan *et al.* [28] think that different pixels in the spatial dimension of HSI should be multiplied by different weights to ensure different smoothness of different pixels. But it ignored the fact that different bands of spectral dimension also need different degrees of processing. One of the important properties of HSIs compared to ordinary images is that HSIs have many different spectral bands. Therefore, we take this into account and modify the spatial dimension adaptive parameter proposed in [28] so that it can be applied to the adaptive processing of spectral dimension. The weight parameters set in this article are different for each spectral band. We set the weight parameter matrix as $W \in \mathbb{R}^{M\times N\times B}$. For the same spectral band, the weight parameters of all pixels are the same. Setting W to the same size as HSI can facilitate the solution after W vectorization.

The weight parameters of all pixels in the ith band are

$$\tau_i = \frac{1}{1+\alpha \sum_{\mathrm{MN}} \|\mathbf{f}\|_{\mathrm{TV}_{2,i}}} \tag{7}$$

where $\|\mathbf{f}\|_{\mathrm{TV}_{2,i}}$ represents the cubic total variation norm of the element of vector $\mathbf{f}$ corresponding to the ith band, while $\sum_{\mathrm{MN}} \|\mathbf{f}\|_{\mathrm{TV}_{2,i}}$ represents the sum operation of the norm of all the elements of vector $\mathbf{f}$ corresponding to the ith band.

In (7), to define the spectrally weighted parameter of each band, we use parameter α to control the contribution of each spectral band information to the adaptive parameter weight τ_i, the range of parameter τ is between [0, 1]. In addition, due to the small value of τ, the iterative solution process of the subsequent augmented Lagrange algorithm [42] is too fast, resulting in poor feature extraction effect. Therefore, a correction value is needed to balance the iteration rate. Through the experiment, the correction value of (NB/M) is the best. When other values are taken, the subsequent classification accuracy is very low, so the correction value is not discussed in the parameter analysis part.

Therefore, the final weight parameters are as follows:

$$[W]_i = \frac{\mathrm{NB}\tau_i}{M}. \tag{8}$$

For the sake of distinction, $[W]_i$ does not refer to the value of the ith parameter, but to the value of the ith bands of the weight matrix W.

Consistent with the representation of $\mathbf{f}$, we use the bold letter $\mathbf{W}$ to represent the vectorized version of matrix W, that is,

$$\mathbf{W} = \mathbf{vec}(W(x, y, z)) \tag{9}$$

where $\mathbf{W} \in \mathbb{R}^{\mathrm{MNB}\times 1}$. And W_i is the ith element of vector $\mathbf{W}$.

Such parameter definition is based on facts. First, the cubic total variation value of a band is positively correlated with the edge contour of the band. When the edge of a band is more obvious, the corresponding weight parameter is smaller, which can avoid serious distortion caused by excessive smoothing.

2) ACTVSP Model: In this work, an adaptive feature extraction smoothing model is proposed. The core optimization problem we solve is to minimize the following formula:

$$\arg\min_{\mathbf{f}} \quad \frac{1}{2}\left\|\mathbf{Hf}-\mathbf{g}\right\|_2^2 + \lambda\, \mathrm{ACTVSP}(\mathbf{f}) \tag{10}$$

where λ is the regularization parameter, which controls the tradeoff between the data fidelity and regularization item, and the regularization item is expressed as

$$\mathrm{ACTVSP}(\mathbf{f}) = \mathbf{W}^T \|\mathbf{f}\|_{\mathrm{TV}_2}. \tag{11}$$

For the convenience of description, refer to the treatment method in [42], we define $\|\mathbf{Df}\|_2 \overset{\mathrm{def}}{=} \|\mathbf{f}\|_{\mathrm{TV}_2}$. Even though they are not exactly equal, we did this to make the description of the algorithm clearer.

Thus, problem (10) can be written as

$$\arg\min_{\mathbf{f}} \quad \frac{1}{2}\left\|\mathbf{Hf}-\mathbf{g}\right\|_2^2 + \lambda \mathbf{W}^T \|\mathbf{Df}\|_2. \tag{12}$$

To solve problem (12), we introduce intermediate variables $\mathbf{u}$ to replace $\mathbf{Df}$. In this way, problem (12) can be transformed into an equivalent problem

$$\begin{aligned} \arg\min_{\mathbf{f},\mathbf{u}} \quad & \frac{1}{2}\left\|\mathbf{Hf}-\mathbf{g}\right\|_2^2 + \lambda \mathbf{W}^T \|\mathbf{u}\|_2 \\ \text{s.t.} \quad & \mathbf{u} = \mathbf{Df}. \end{aligned} \tag{13}$$

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Therefore, the augmented Lagrange function to solve the problem (13) is

$$L(\mathbf{f},\mathbf{u},\mathbf{y}) = \frac{1}{2}\left\|\mathbf{Hf}-\mathbf{g}\right\|_2^2 + \lambda\mathbf{W}^T\|\mathbf{u}\|_2 - \mathbf{y}^T(\mathbf{u}-\mathbf{Df}) + \frac{\rho}{2}\left\|\mathbf{u}-\mathbf{Df}\right\|_2^2 \quad (14)$$

where ρ is a fixed parameter representing the step size of gradient ascent, $\|\mathbf{u}-\mathbf{Df}\|_2^2$ is quadratic penalty term, and $\mathbf{y}$ is the Lagrange multiplier associated with the constraint $\mathbf{u}=\mathbf{Df}$.

Then, through continuous iteration, the saddle point of the Lagrange function $L(\mathbf{f},\mathbf{u},\mathbf{y})$ is finally found, that is, the solution of the original problem (10). Referring to [42] of video denoising solutions, we use the alternating direction method (ADM) [43], [44] to iteratively solve $\mathbf{f}$ and $\mathbf{u}$ subproblems iteratively, which results in the following updates.

Step 1: Update $\mathbf{f}_{k+1}$

$$\mathbf{f}_{k+1} = \arg\min_{\mathbf{f}} \frac{1}{2}\left\|\mathbf{Hf}-\mathbf{g}\right\|_2^2 - \mathbf{y}_k^T(\mathbf{u}_k-\mathbf{Df}) + \frac{\rho}{2}\left\|\mathbf{u}_k-\mathbf{Df}\right\|_2^2. \quad (15)$$

By dropping the indices k, and taking the partial with respect to $\mathbf{f}$, solution of the subproblem (15) is found by considering the normal equation

$$\mathbf{H}^T\mathbf{Hf} - \mathbf{H}^T g + \mathbf{D}^T\mathbf{y} - \rho\mathbf{D}^T\mathbf{u} + \rho\mathbf{D}^T\mathbf{Df} = 0. \quad (16)$$

By arranging (16), we can get

$$(\mathbf{H}^T\mathbf{H} + \rho\mathbf{D}^T\mathbf{D})\mathbf{f} = \mathbf{H}^T\mathbf{g} + \rho\mathbf{D}^T\mathbf{u} - \mathbf{D}^T\mathbf{y} \quad (17)$$

after the triple block-circulant convolution matrix $\mathbf{H}$ be diagonalized by the 3-D DFT matrix, one solution of (17) is

$$\mathbf{f} = \mathscr{F}^{-1}\left[\frac{\mathscr{F}\left(\mathbf{H}^T\mathbf{g} + \rho\mathbf{D}^T\mathbf{u} - \mathbf{D}^T\mathbf{y}\right)}{\left|\mathscr{F}[\mathbf{H}]\right|^2 + \rho\left(\left|\mathscr{F}[\mathbf{D}_x]\right|^2 + \left|\mathscr{F}[\mathbf{D}_y]\right|^2 + \left|\mathscr{F}[\mathbf{D}_z]\right|^2\right)}\right] \quad (18)$$

where $\mathscr{F}$ denotes the 3-D Fourier transform operator [45]–[47].

Step 2: Update $\mathbf{u}_{k+1}$

$$u_{k+1} = \arg\min_{\mathbf{u}} \lambda\mathbf{W}^T\|\mathbf{u}\|_2 - \mathbf{y}_k^T(\mathbf{u}-\mathbf{Df}_{k+1}) + \frac{\rho}{2}\left\|\mathbf{u}-\mathbf{Df}_{k+1}\right\|_2^2. \quad (19)$$

The $\mathbf{u}$-subproblem can be solved using the shrinkage formula [48]. Letting $\mathbf{v} = \mathbf{Df} + (\mathbf{y}/\rho)$, $\mathbf{u}$ is given by the following formula:

$$\mathbf{u} = \max\left\{\|\mathbf{v}\|_2 - \frac{\lambda\mathbf{W}^T}{\rho}, 0\right\} \cdot \frac{\lambda\mathbf{W}^T\mathbf{v}}{\|\mathbf{v}\|_2}. \quad (20)$$

We can easily find that

$$\mathbf{u}_x = \max\left\{\|\mathbf{v}\|_2 - \frac{\lambda\mathbf{W}^T}{\rho}, 0\right\} \cdot \frac{\lambda\mathbf{W}^T\mathbf{v}_x}{\|\mathbf{v}\|_2} \quad (21)$$

where $\mathbf{v}_x = \beta_x\mathbf{D}_x\mathbf{f} + (\mathbf{y}_x/\rho)$, $\mathbf{v}_y$, and $\mathbf{v}_z$ can be derived in a similar way.

Step 3: Update $\mathbf{y}_{k+1}$

$$\mathbf{y}_{k+1} = \mathbf{y}_k - \rho(\mathbf{u}_{k+1} - \mathbf{Df}_{k+1}). \quad (22)$$

Repeat these three steps until they converge at each pixel. From this, we can get the image $\mathbf{f}$ after feature extraction. To further enhance the separability of pixels between different species, the kernel principal component analysis (KPCA) [49] is applied on $\mathbf{f}$ obtained by the above steps, and K principal components are preserved for classification.

Then, the SVM classifier [8] is conducted to obtain the class probability $\mathbf{P}_1$.

3) Edge-Preserving Feature Extraction: Through the first stage of image smoothing and feature extraction, the HSI obtained will improve the classification effect, but then face the problem that the image edge contour becomes blurred after texture smoothing. At the same time, because of too many parameters, it is difficult to debug the best parameter combination in practical application. Edge-preserving filtering [39] can make up for the above problems in the first stage and enhance the robustness of the overall algorithm while sharpening the edges of the contour to prevent excessive smoothing. To be more specific, the dimensionally reduced images are first fed into a spectral classifier SVM [8] to obtain multiple initial probability maps. For example, $\mathbf{p} = (\mathbf{p}_1, \ldots, \mathbf{p}_n)$ represents one of the probability maps, where $\mathbf{p}_{i,n} \in [0,1]$ is the initial probability that pixel i belongs to the nth class.

Then edge-preserving filtering is used to optimize the obtained probability map and align the probability map with real object boundaries. The optimization probability is modeled as the weighted average of its neighborhood probability, that is,

$$\acute{\mathbf{p}}_{i,n} = \sum_j W_{i,j}(I)\mathbf{p}_{j,n} \quad (23)$$

where i and j represent the ith and jth pixel points, respectively, and the filtering weight W is selected so that the filtering retains the edge of the specified guiding image I. The guided filter is used in this article, and the filtering weight $W_{i,j}(I)$ of the guided filter can be expressed as follows:

$$W_{i,j}(I) = \left(\frac{1}{|\omega|^2}\right)\sum_{k\in\omega_i, k\in\omega_j}\left(1 + \frac{(I_i - m_k)(I_j - m_k)}{\sigma_k^2 + \epsilon}\right) \quad (24)$$

where ω_i and ω_j are local windows around pixel i and j, respectively, m_k and σ_k are the mean and variance of I in ω_k, and $|\omega|$ is the number of pixels in ω_k. Gray-scale guidance image obtained after principal component analysis is used as guidance image I.

Finally, a classification based on maximum probability is used to obtain the final probability, according to the following equation:

$$\acute{\mathbf{p}}_i = \arg\max_n \acute{\mathbf{p}}_{i,n}. \quad (25)$$

The final $\acute{\mathbf{p}}$ is $\mathbf{P}_2$ of the second stage probability map we need.

4) Probability Fusion: The class probability obtained through the first stage may cause the edges to be blurred compared with the original image due to excessive smoothing of the image. In the second stage, the class probability is obtained by using edge-preserving filtering [39] algorithm, and the edge is relatively clear. In this article, weighted decision

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fusion rule is used to merge the class probability obtained in the two stages, so as to obtain the final class label. In this way, the classification probability of the same target is not only higher than that of other types of land cover but also the edge will be very clear, which can more accurately divide the area covered by different ground objects. In terms of pixels, the final label for each pixel is determined based on maximum probability

$$\mathbf{P} = \arg\max_i \left\{ \mu \mathbf{P}_1^i + (1-\mu)\mathbf{P}_2^i \right\} \tag{26}$$

where μ is a free parameter, and $\mathbf{P}_1^i$ is the class probability map obtained by the classification of ACTVSP. $\mathbf{P}_2^i$ denotes the class probability map of classification of EPF. $\mathbf{P}$ is the final classification result.

The complete and detailed algorithm is given in Algorithm 1 for easy understanding.

Algorithm 1 Algorithm Process

Input observed HSI $\mathbf{g}$
step1.**ACTVSP**
Initialize $\mathbf{f_0} = \mathbf{g}, \mathbf{u_0} = \mathbf{Df_0}, \mathbf{y} = 0, k = 0$
while $\|\mathbf{f}_{k+1} - \mathbf{f}_k\| > \text{tol}$ do
  Calculate $\mathbf{W}$.
  Solve the f-subproblem using Equation (18).
  Solve the u-subproblem using Equation (20).
  Update the Lagrange multiplier y using Equation (22).
end
  Get probability map $\mathbf{P_1}$ via SVM.
step2.**EPF**
  Obtain multiple initial probability maps.
  Optimize and align the obtained probability map by Equation (23).
  Get probability map $\mathbf{P_2}$ using Equation (25).
step3.**Probability Fusion**
  Get final probability map $\mathbf{P}$ using Equation (26).

C. Parameter Optimization and Selection of SVM Model

In this article, SVM is selected as the classifier for both pre-processing method ACTVSP and post-processing method EPF.

For the SVM with Gaussian kernel function, there are two main training parameters: C is the penalty coefficient, which controls over-fitting of the model, and gamma (γ) is a parameter that comes with the kernel function, which controls the degree of nonlinearity of the model.

For the determination of C and gamma parameters, we use the cross-validation method and the grid search algorithm as the parameter optimization method. The C and gamma parameters are combined according to step size 1 in a certain range of values. (The certain value range is pointed out in Section IV-B2.) Then, under different (C, gamma) combinations, the training set samples are divided into n groups. One group is used as validation data samples, and the other $n-1$ groups are used as training data. Each set of data is taken in turn as a verification sample, so that under a combination of (C, Gamma), n calculations are required. The n-times model accuracy was averaged as a score for the models in this group (C, Gamma).

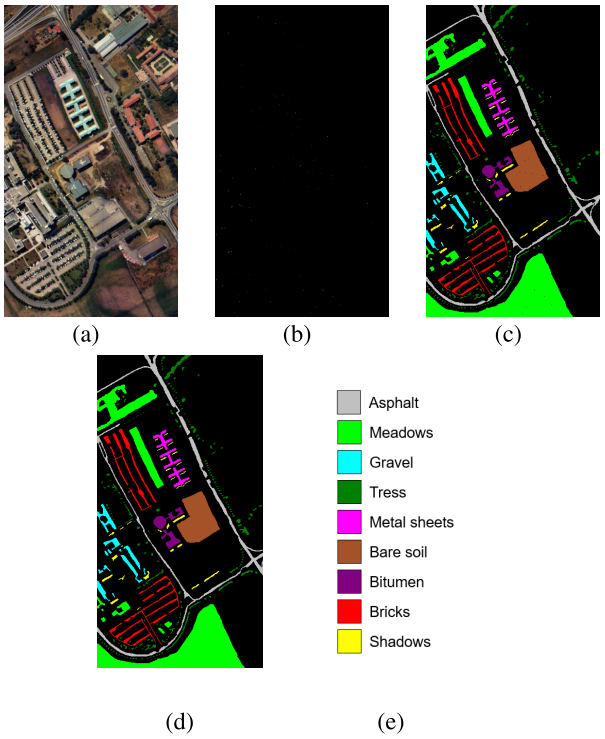

Fig. 2. University of Pavia dataset. (a) False color composite image. (b) Training samples. (c) Testing samples. (d) Ground truth. (e) Class names.

In this way, we can get the score of the model under different (C, gamma) combinations. We choose the group with the highest score (C, gamma) as the final parameter value of SVM model. If there are different combinations that have the same score, we first consider the parameter C, the smaller value is prior. Because under the condition that the accuracy of the model is guaranteed, the smaller the value of C is, the higher the fault-tolerance of the model is. Thus, we can avoid over-fitting.

The above parameter optimization process can make the parameters of SVM model change dynamically with the training set, and avoid the tedious manual adjustment process.

IV. Experiments

A. Datasets

To verify the validity of our method, we use a classical dataset, i.e., University of Pavia, a dataset used for hyperspectral change detection in the past, i.e., the China Dataset, and a recent dataset, i.e., Houston 2013.

1) University of Pavia: The University of Pavia image was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS-03) optical sensor with a spectral coverage ranging from 0.43 to 0.86 μm, which covers the campus at the University of Pavia, Pavia, Italy. This image contains 103 bands of size 610 $\times$ 340 pixels with a spatial resolution 1.3 m per pixel. Fig. 2 shows the false color composite image, training and test samples, and ground truth. The number of training and test samples is shown in Table VI.

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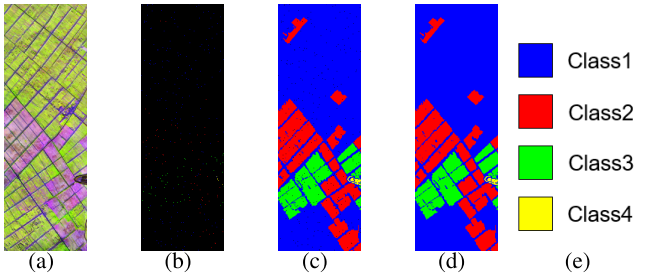

Fig. 3. China dataset. (a) False color composite image. (b) Training samples. (c) Testing samples. (d) Ground truth. (e) Class names.

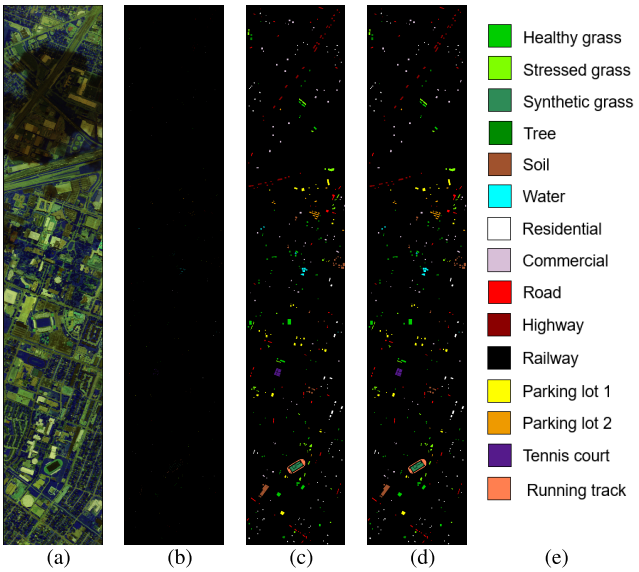

Fig. 4. Houston 2013 dataset. (a) False color composite image. (b) Training samples. (c) Testing samples. (d) Ground truth. (e) Class names.

2) China Dataset: The China dataset belongs to a farmland near the city of Yuncheng Jiangsu province in China, which was captured for hyperspectral change detection on May 3, 2006, and April 23, 2007 [50]. This scene is mainly a combination of soil, river, tree, building, road, and agricultural field. For this dataset, all changes related to the type of land cover and river. And the ground truth provided by the author was divided into four main categories. In addition, this dataset belongs to Hyperion sensors. This image contains 154 bands of size 420 × 140 pixels. Fig. 3 shows the false color composite image, training and test samples, and ground truth. The number of training and test samples is shown in Table VII.

3) Houston 2013: The Houston 2013 image was distributed by the 2013 GRSS Data Fusion Contest. This image was acquired by the compact airborne spectrographic imager over the University of Houston campus and the neighboring urban area on June 23, 2012. Furthermore, it contains 144 bands of size 349 × 1905 pixels with a spatial resolution of 2.5 m, and the spectral coverage ranges from 0.38 to 1.05 μm. Fig. 4 shows the false color composite image, training and test samples, and ground truth. The number of training and test samples is shown in Table VIII.

B. Experimental Setup

1) Evaluation Indexes: In order to quantitatively evaluate the classification performance of all research methods, four popular and widely used quantitative indicators, i.e., individual class accuracy (CA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient, were used to evaluate the classification performance of HSI. Where CA represents the percentage of correctly classified pixels for each class. OA represents the percentage of correctly classified pixels in the total number of pixels. AA measures the average of the percentage of pixels correctly identified for each land cover. The Kappa coefficient calculates the percentage of identified pixels corrected by the number of agreements that would be expected purely by chance. All experiments were randomly selected and repeated for ten times to obtain the mean and standard deviation of CA, OA, AA, and Kappa coefficients. The final result is obtained by taking the average value of the four evaluation indexes obtained from the above experiments.

2) Competitive Approaches: In this article, the following seven state-of-the-art classification methods are used for comparison, including four machine learning methods and three deep learning methods:

SVM [8] is performed using the LIBSVM library, where the Gaussian kernel is adopted. Among them, the optimal parameters selection of kernel adopts fivefold cross-validation. The penalty factor is from 10^{-2} to 10^{4}, and the kernel width varies between $2^{-4} \sim 2^{4}$.

Weighted Markov Random Field (WMRF) [26] uses the total variational regularization model to smooth the a posteriori distribution to describe the spatial information in the hidden domain.

Random Patches Network (RPNet) [51] is fed into a spectral classifier using random patch as a convolution kernel by learning the deep convolution features of HSI.

Structural Profile (SP) [38] proposes an adaptive texture smoothing method to construct the structural contour of HSIs and a HSI classification framework based on dual spatial information fusion.

Random Multigraphs Ensemble Learning (RMGE) [52] propose a novel graph-based semi-supervised learning classification model for HSIs.

Spectralformer [53] rethink HSI classification from a sequential perspective with transformers and propose a novel backbone network called SpectralFormer for HSI classification.

Weighted Feature Fusion (WFCG) [54] propose a weighted feature fusion framework called WFCG for HSI classification, by using the characteristics of superpixel-based GAT and pixel-based CNN, which proved to be complementary.

All comparison methods used the default parameter settings given in the corresponding literature for comparison.

C. Analysis of Parameters

In the proposed method, there are six parameters need to be fixed, i.e., the number of the dimension reduced data B, the constants $(\beta_x, \beta_y, \beta_z)$, the number of kernel principle components K used for classification, the regularization parameter λ, the adaptive parameter contribution coefficient α, and the

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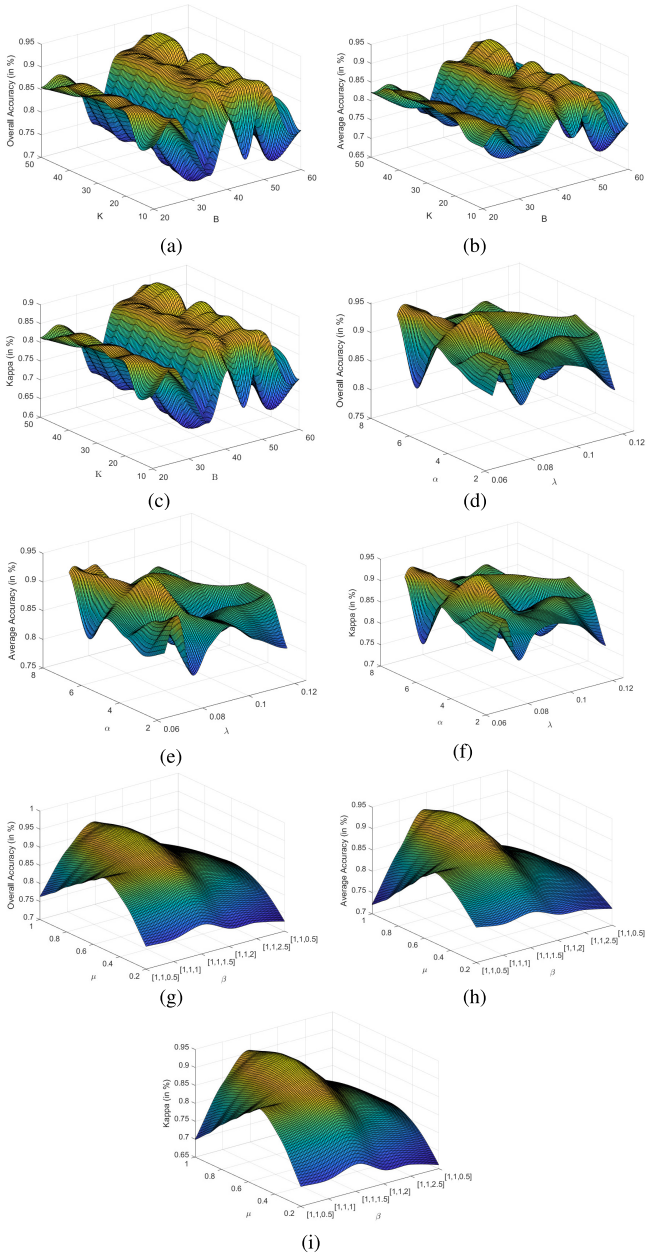

Fig. 5. Influence of the parameters, i.e., B, K, λ, α, μ, and $(\beta_x, \beta_y, \beta_z)$, on the classification performance of the proposed method (a) OA. (b) AA. (c) Kappa. (d) OA. (e) AA. (f) Kappa. (g) OA. (h) AA. (i) Kappa.

fusion weight μ. An experiment is performed on the University of Pavia dataset to initialize the aforementioned parameters. When the parameters B and K are analyzed, $(\beta_x, \beta_y, \beta_z)$, λ, α, and μ are set to be (1,1,1), 0.083, 6, and 0.7, respectively. Fig. 5(a)–(c) shows the influence of different values of B and K on OA, AA, Kappa, and computing time. It can be concluded that, in general, when B and K are smaller, the classification accuracy of our method tends to decline and the calculation time is shorter. Moreover, when B and K are set to 50 and 45, respectively, OA and AA of the method reach the best classification accuracy, and Kappa coefficient is also large.

Besides, based on the most suitable values of B and K obtained from the above experiments (i.e., B and K are set as 50 and 45, respectively), Fig. 5(d) to (f) shows the influence of parameters λ and α. Obviously, when α is equal to 5 and λ is around 0.083, OA, AA, and Kappa can achieve the best results. Fig. 5(g)–(i) shows the influence of parameters μ and β when other parameters are set to the most appropriate value obtained by experiments (B, K, λ and α are set as $50, 45, 0.083$, and 5, respectively). It can be concluded that, when β is equal to [1,1,1] and μ is equal to 0.7, the three evaluation indexes achieved the best effect. Subsequent experiments based on optimal parameters verify that the proposed method can achieve good classification performance on all datasets.

TABLE I
CLASSIFICATION EFFECT OF DIFFERENT DATA DIMENSION

Data type	Raw data	Dimension reduced data					
		20	30	40	50	60	70
OA (in %)	88.31	84.23	88.36	90.65	**95.07**	81.23	81.97
AA (in %)	84.14	80.72	83.91	85.66	**92.91**	81.38	77.11
Kappa (in %)	84.88	79.83	84.81	87.90	**93.52**	76.04	77.16
time (in s)	149.21	**40.58**	50.28	60.61	86.76	98.37	117.62

D. Analysis of the Effects of Different Data Dimension

First of all, the purpose of dimensionality reduction for original data is to reduce run time and redundant information. Table I lists the classification effects of data with original data and different dimensionality reduction values on the University of Pavia dataset by using the method proposed in this article. The number of training samples is 10 pixels randomly selected from each class. As can be seen from Table I, for the data with appropriate values for dimensionality reduction, both the overall accuracy of classification and the running time are significantly improved. For example, on the University of Pavia dataset, the best classification effect can be achieved by sending the data after dimensionality reduction to 50 bands into subsequent work. However, it does not mean that the higher the dimension reduction intensity is, the better the effect of improving the classification accuracy will be. Excessive dimension reduction will lead to the neglect of useful information and the increase of running time. The insufficient dimension reduction will reduce the running time, but it will cause a lot of redundant information to be retained. In both cases, the classification accuracy will decrease. In general, due to the large amount of useless information in HSIs, dimensionality reduction by selecting appropriate values of original data at the beginning can improve the classification effect.

Fig. 6 compares the probability map obtained by using the original data and the data with a dimensionality reduction value of 50. It can be clearly seen that the classification probability map obtained by using the dimensionality reduction data is very similar, and even clearer in detail.

E. Ablation Experiment

The two main components of the proposed method are feature extraction by ACTVSP model and edge-preserving filtering [39]. The first stage can be divided into ACTVSP

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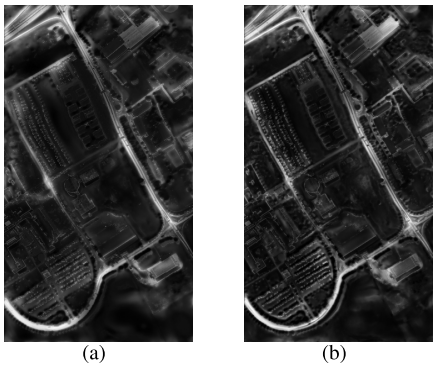

(a) (b)

Fig. 6. Probability maps obtained by the classification of the University of Pavia dataset. (a) Using original image classification. (b) Using dimension reduced image.

TABLE II

CLASSIFICATION EFFECT ON THE UNIVERSITY OF PAVIA DATASET WITH DIFFERENT DIMENSION REDUCTION METHODS

	PCA	LDA	ProbPCA	FA	MDS	KPCA
OA (in %)	90.75	86.13	88.89	63.45	92.57	**95.07**
AA (in %)	86.51	79.19	86.14	69.54	92.14	**92.91**
Kappa (in %)	87.86	81.76	85.73	55.01	90.03	**93.52**
time (in s)	70.51	70.47	317.45	114.11	**68.85**	72.22

feature extraction and dimension reduction using KPCA [49]. Experiments were conducted on the University of Pavia dataset using the control variable method, where the number of training samples was 10 pixels randomly selected from each class. First, we analyzed the influence of different dimension reduction method for classification effect. It should be noted that we used the same best combination of parameters for all the experiments which was acquired in Section IV-C. Table II shows the classification performance obtained by combining several different dimension reduction methods with other components. Specifically, we selected six widely used the method for comparison, i.e., principal component analysis (PCA) [55], linear discriminant analysis (LDA) [56], probabilistic PCA (ProbPCA) [57], factor analysis (FA) [58], classical multidimensional scaling (MDS) [59], and kernel PCA (KPCA) [49]. It can be seen from Table II that the KPCA method combined with other components can obtain the highest classification accuracy. The reason is that KPCA projects the data of the original space into the high-dimensional feature space through kernel mapping for processing.

Furthermore, the results in Table III demonstrate the validity of other different components. First of all, it can be seen from Table III that only using one of EPF, ACTVSP and KPCA for feature extraction, or even the fusion method of ACTVSP and EPF for classification results are unsatisfactory. In addition, it can be observed that classification result acquired by the combination of KPCA and EPF are very poor, which can prove that KPCA does not play a key role in the final classification results. After image smoothing and feature extraction by ACTVSP and dimension reduction by KPCA [49], the final classification effect is greatly improved. On this basis, the classification effect obtained by probability combination with EPF is the best in the experimental method. This can be explained. To illustrate visually, Fig. 7 shows the separability of different categories of pixels after combining them using different methods.

As can be seen from Fig. 7, the separability of the image processed by ACTVSP is improved, because of pixels of the same kind are gathered together, and then KPCA processing [49] is performed to separate pixels of different kinds more widely, the separability of pixels is further improved, thus improving the classification efficiency.

To show more intuitively that our methods can improve separability between categories, we choose several widely used baseline methods [36] in the hyperspectral community for comparison, including Gabor [13], LPP [60], LDA [56], CGDA [61], JPlay [62], and SP [38]. The experiment was performed on the University of Pavia dataset. For the sake of convenience, we set the number of spectral dimensions after the feature extraction to 45. Fig. 8 shows the scatter diagrams of the first three bands, which can represent class separation capability of features extracted by different methods. As we all know, the higher the degree of separation between classes, the better the classification result obtained by the classifier.

It can be observed from Fig. 8, compared with other single feature extraction methods, SP and our method can effectively reduce intraclass differences and increase interclass differences. Because these two methods smooth the HSI by adding the variational regularization term, and retain the main features. However, our method further widens the interclass gap and further reduces the intraclass gap compared with SP. This can be explained because our method adds smoothing to the spectral segment during the pre-processing stage, which removes as much redundant information as possible and makes the main features clearer.

F. Description of Robustness and Smoothing Effect

To prove the robustness and generalizability of our method, we conducted experiments on ten public datasets, including Indian Pines 1992 (220 bands version), the University of Pavia, Pavia Center Part1, Pavia Center Part2, Salinas, the Kennedy Space Center, the China dataset, Indian Pines 2010, Botswana and Houston 2013 dataset. All datasets were randomly selected from 20 training samples in each category and the rest as test samples. In addition, all datasets were experimented using the same set of fixed parameters (B, K, λ, α, μ, and (β_x, β_y, β_z) equal to 50, 45, 0.083, 5, 12 and [1,1,1], respectively). Among all datasets, some differ greatly in spatial size, for example, Indian Pines 1992 contains 145×145 pixels, while Houston contains 349×1905 pixels. Some differ greatly in spectral dimensions, for example, Indian Pines 1992 contains 220 bands, while the University of Pavia contains 103 bands. In addition, the proportion of different classes in almost all datasets also varies greatly. For example, the number of four categories in the China dataset are 40 407, 12 229, 6019, and 145. Therefore, taking 20 random points as training samples for each class will bring great challenges to classification. Table IV shows the classification results. It can be observed that, despite many of the problems mentioned above, our approach achieves over 85% overall accuracies

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TABLE III
CLASSIFICATION EFFECT OF DIFFERENT COMPONENTS

	EPF	ACTVSP	KPCA	KPCA+EPF	ACTVSP+KPCA	ACTVSP+EPF	ACTVSP+KPCA+EPF
OA (in %)	66.35	65.45	46.18	70.88	92.86	74.03	**95.07**
AA (in %)	67.76	68.69	53.86	76.89	90.92	69.02	**92.91**
Kappa (in %)	58.62	57.77	36.32	62.75	90.60	67.88	**93.52**
time (in s)	**0.40**	72.64	5.51	0.41	70.69	73.53	72.22

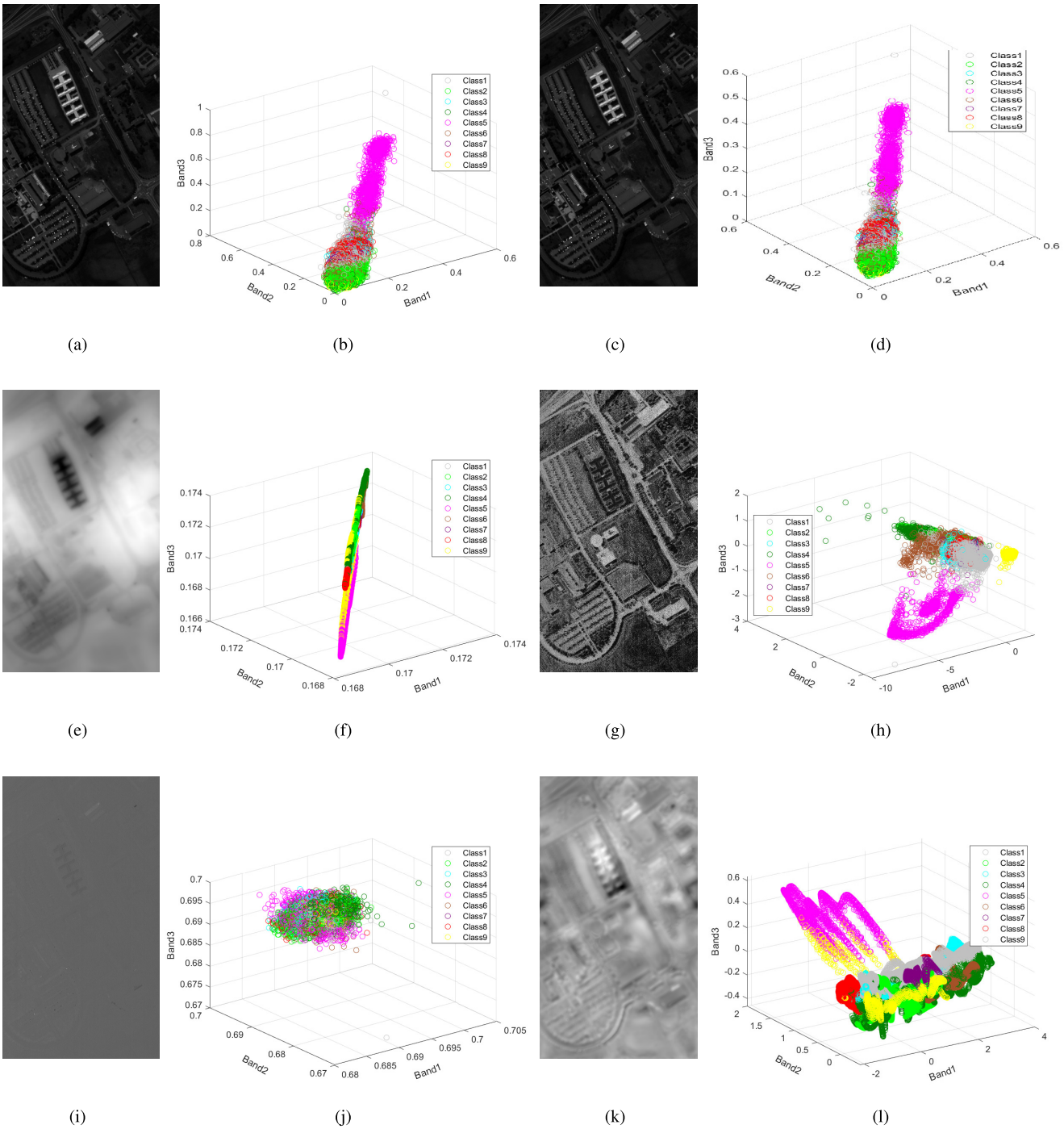

Fig. 7. Influence of different components on pixel separability of the University of Pavia dataset. (a) Dimensionality reduced image. (b) Pixel separability of the first three bands about (a) (different symbols denotes pixels belonging to different classes). (c) EPF processed image. (d) Pixel separability after EPF. (e) Images extracted by ACTVSP. (f) Pixel separability after ACTVSP. (g) KPCA processed image. (h) Pixel separability after KPCA. (i) KPCA+EPF processed image. (j) Pixel separability after KPCA+EPF. (k) ACTVSP+KPCA processed image. (l) Pixel separability after ACTVSP+KPCA.

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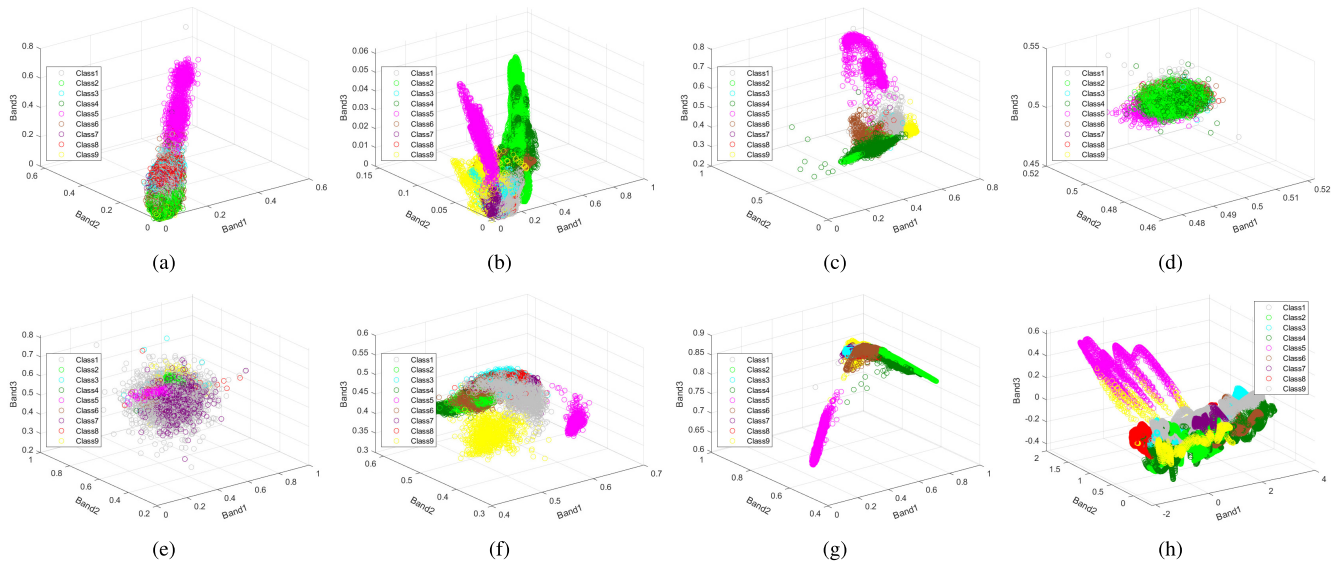

Fig. 8. Pixel separability of baseline methods. (a) Original HSI. (b) Gabor [13]. (c) LPP [60]. (d) LDA [56]. (e) CGDA [61]. (f) JPlay [62]. (g) SP [38]. (h) Our method.

TABLE IV
CLASSIFICATION PERFORMANCE ON TEN PUBLIC DATASETS

	Indian Pines 1992	Pavia University	Pavia Center Part1	Pavia Center Part2	Salinas	KSC	China	Indian Pines 2010	Botswana	Houston 2013
OA (in %)	91.08	96.29	97.46	97.46	97.92	99.07	85.46	96.30	99.19	85.84
AA (in %)	87.40	94.40	92.33	92.33	98.32	98.81	68.66	95.09	99.30	86.78
Kappa (in %)	89.82	95.12	96.42	96.42	97.68	98.96	70.88	95.46	99.12	84.70
time (in s)	13.57	88.64	279.08	284.88	37.07	122.26	14.63	137.58	116.32	225.09

on all experimental datasets, and over 90% on eight datasets. Although there are six parameters listed in Section IV-C that need to be adjusted, but the purpose of adjustment is to achieve the best classification effect. In practice, we do not need to spend a lot of energy on parameter selection, but only need to select a set of fixed universally applicable parameters can achieve better results. To get the best results, we just need to make a few minor adjustments. This is why we say that the proposed method is of robustness.

Moreover, the objective evaluation of smoothing effect on band1, band2, band3, and band4 are shown in Table V. At the same time, for comparison, we also quantitatively evaluate the smoothing effect of SP method on the above four bands. It can be concluded that the proposed method does have a certain smoothing effect with regard to quantitative measurements aspect. Although our method uses an adaptive smoothing strategy for spectral dimensions, the SP method smoothens HSIs in spatial dimensions, but the quantitative evaluation shows that the smoothing effect is similar for HSIs. This further illustrates that rich spectral bands are a very important property of HSIs. In recent years, many methods of HSI classification focus on the processing of spatial information rather than making full use of spectral information. Our approach is to regress to the essence and to introduce spatial information on the basis of spectral information. It turns out that this can also obtain a good feature extraction effect, and then improve the classification effect. In addition, it can be seen from Figs. 9(d), 10(d) and 11(d) that the classification map

TABLE V
OBJECTIVE EVALUATION OF SMOOTHING EFFECT ON DIFFERENT BANDS

Method	Evaluation Index	Band1	Band2	Band3	Band4
Our method	MSE	1.831026	**0.080332**	**0.047911**	0.038629
	PSNR	45.50386	**59.08193**	**61.32643**	62.26171
SP	MSE	**1.695135**	0.221813	0.148387	**0.026773**
	PSNR	**45.83876**	54.67094	56.41684	**63.85387**

obtained by SP has over-fitting phenomenon, which is caused by the excessive smoothing of HSIs in the pre-processing stage. Therefore, although our method and SP method are very close to each other in the evaluation index of denoising, SP method will cause more information loss.

G. Classification Results

1) University of Pavia: The first experiment was conducted on the University of Pavia dataset. Among them, 10 pixels are randomly selected from each class for the training samples, accounting for 0.2% of ground truth. In all contrast methods, SVM [8], WMRF [26], SP [38], and RMGE [52] are traditional machine learning methods. RPNet [51], Spectralformer [53], and WFCG [54] are deep learning methods.

It can be clearly seen from Fig. 9 that many noise labels are misclassified in the visual ground object classification map of SVM [8]. The reason is that SVM does not process original HSI and only uses spectral information while ignoring spatial information, which leads to unsatisfactory overall

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TABLE VI

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR THE UNIVERSITY OF PAVIA DATASET, INCLUDING SVM [8], WMRF [26], RPNET [51], SP [38], RMGE [52], SPECTRALFORMER [53], WFCG [54], AND OUR METHOD

Class name	Training set	Testing set	Classification accuracies of different approaches (in %)							
			SVM	WMRF	RPNet	SP	RMGE	Spectralformer	WFCG	Our method
Asphalt	10	6621	88.07	67.54	97.16	**97.27**	65.81	72.56	95.92	96.73
Meadows	10	18639	91.34	76.97	97.07	97.58	93.03	50.43	**99.10**	98.99
Gravel	10	2089	55.82	68.60	64.72	**99.14**	68.70	36.38	93.03	77.92
Tress	10	3054	63.85	90.93	94.04	76.24	28.82	94.07	**99.09**	76.68
Metal sheets	10	1335	90.77	99.33	63.04	79.13	**99.93**	99.10	89.77	97.23
Bare soil	10	5019	32.21	62.74	47.49	99.27	66.10	69.08	99.76	**99.78**
Bitumen	10	1320	48.44	68.48	57.94	**100.00**	73.46	46.52	86.49	91.32
Bricks	10	3672	63.15	55.07	82.42	82.99	44.98	77.26	71.49	**97.61**
Shadows	10	937	99.89	43.54	99.43	99.64	79.94	**100.00**	99.81	99.89
	OA		64.41	72.24	79.66	94.22	75.03	63.29	94.57	**95.07**
	AA		70.39	70.35	78.15	92.36	68.97	71.71	92.72	**92.91**
	Kappa		57.10	64.39	74.37	92.34	65.42	55.01	92.77	**93.52**

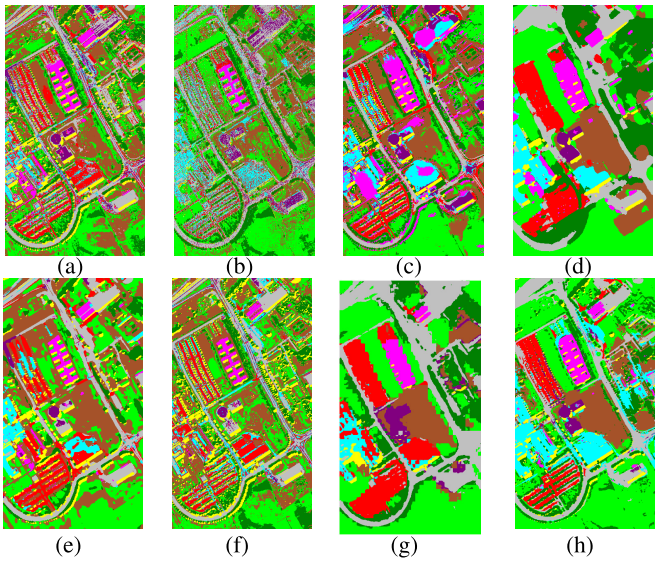

Fig. 9. Classification maps of all studied approaches on the University of Pavia dataset. (a) SVM [8], (b) WMRF [26], (c) RPNet [51], (d) SP [38], (e) RMGE [52], (f) Spectralformer [53], (g) WFCG [54], and (h) Our method.

accuracy of classification. The WMRF method [26] uses the spatial adaptive total variational regularization method to make the classification more smooth in space by smoothing HSI images, so as to improve the classification accuracy. However, there are many mislabeled pixels in this classification map, because when the number of training samples is limited, the classification map obtained by WMRF method will appear excessive smoothness. RPNet [51], SP [38] use the dual information of spatial and spectral to reduce the influence of noise on classification results. Compared with SVM, RPNet, and SP have better classification effect. RMGE [52] also uses spatial and spectral information. Even though it retains the fine structure of the local region by modeling multiple anchor graphs, this semi-supervised approach of representing the whole by parts is not as good as methods which deal with each spectral segment. Besides, due to limited training samples, Spectralformer [53] have large areas of mislabeled pixels. Among the three deep methods, WFCG [54] performs best, because it is a feature fusion method that use the characteristics of superpixel-based GAT and pixel-based CNN. However, compared with our method, it needs more training samples, so the classification effect is slightly worse than our method. SP method has better robustness compared with other methods due to its fusion framework. However, due to the large number of parameters, unreasonable parameter setting will lead to over-fitting phenomenon, small samples will be ignored, and excessive image smoothing will easily lead to misclassification. For example, in Fig. 9(d), small targets within the main distribution areas of meadows, metal sheets, and bricks are almost all smoothed out. In contrast, the proposed method combines spectral and spatial information, draws on the noise robustness characteristics of SP framework, and proposes a complementary method, which can extract objects while preserving object edges. Compared with other methods, the classification accuracy is higher when the ground object scene is complicated, the edge is not obvious and the distribution is scattered.

Quantitative results obtained by different research methods are shown in Table VI. As can be seen from the table, our method has the highest classification accuracy in OA, AA, and Kappa coefficients. In addition, although only three classes of our method have the highest CA, six of the nine classes have CA of more than 95%, and seven of them have CA of more than 90%. This shows that compared with other comparison methods, the method presented in this article is widely applicable to the classification of most different types of land types on the University of Pavia dataset.

2) China Dataset: The second experiment was conducted on the China Dataset. Since the total number of samples of different categories is too large, we randomly select 200, 70, 50, and 5 labeled points as training samples for four categories, accounting for 0.49%, 0.57%, 0.84%, and 3.6% of ground truth, respectively. In addition to class 4, the interclass similarities of the other three classes are very strong. The classification maps of all methods are shown in Fig. 10.

Through observation, due to the small number of training samples and the small proportion in the image of class 4, the classification effect of all the methods on class 4 is the

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TABLE VII

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR THE CHINA DATASET, INCLUDING SVM [8], WMRF [26], RPNET [51], SP [38], RMGE [52], SPECTRALFORMER [53], WFCG [54], AND OUR METHOD

Class name	Training set	Testing set	Classification accuracies of different approaches (in %)							
			SVM	WMRF	RPNet	SP	RMGE	Spectralformer	WFCG	Our method
class1	200	40207	88.45	90.67	91.77	92.02	92.60	89.98	90.22	**93.24**
class2	70	12159	86.40	86.49	86.95	88.02	84.59	80.64	90.94	**93.40**
class3	50	5969	68.31	68.40	71.30	91.50	73.70	46.51	**94.60**	82.66
class4	5	140	13.33	57.14	51.61	56.04	**79.31**	46.51	0.00	61.74
	OA		85.35	84.37	88.78	91.01	88.96	83.51	90.65	**92.14**
	AA		69.54	75.68	75.41	81.90	82.55	67.14	68.94	**82.76**
	Kappa		67.55	67.51	76.04	80.42	76.53	64.66	78.96	**83.10**

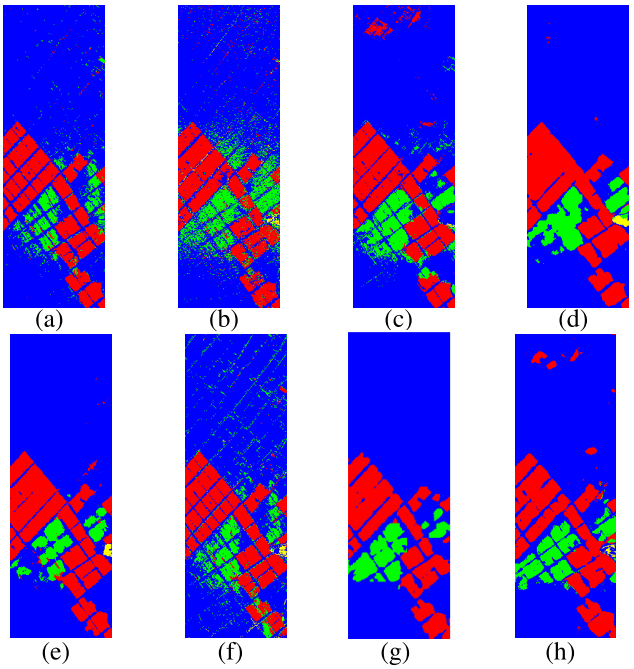

Fig. 10. Classification maps of all studied approaches on the China dataset. (a) SVM [8], (b) WMRF [26], (c) RPNet [51], (d) SP [38], (e) RMGE [52], (f) Spectralformer [53], (g) WFCG [54], and (h) Our method.

worst. Among them, the three deep methods (RPNet [51], Spectralformer [53], and WFCG [54]) for class 4 misclassification phenomenon is more obvious, WFCG method even completely misclassified class 4. One important reason is that deep methods generally require more training samples than traditional machine learning methods. Besides, it can be concluded that since SVM [8] only uses spectral information as the classification basis, the obtained classification map is also greatly affected by noise, and the misclassification phenomenon is serious. The same goes for WMRF [26]. Different from the excessive smoothness of the University of Pavia dataset, WMRF has the problem of insufficient smoothness on the China dataset, which leads to the phenomenon of misclassification of noise labels. Since class 1 is similar to class 2, all the methods except our method and RPNet did not classify class 2 in the upper left corner correctly, but misclassified class 2 in the upper left corner into class 1. However, because of the similarity between class 1 and class 3, and the interference of noise, RPNet, as a deep method, produced serious misclassification when the training samples are limited.

It can be seen from the classification maps that SP [38] and WFCG have good classification effects. An important reason is that these methods all make use of the dual information of spectral and spatial. Among them, the classification map of SP has the phenomenon of over-fitting, which is because the feature extraction process of SP is a denoising process in essence, leads to over-smoothing and insufficient extraction of edges and details. Compared to the University of Pavia dataset, the China dataset has a high degree of interclass similarity, so it is a big challenge to distinguish between different classes. Meanwhile, due to the increase in the proportion of different classes, it is easy to ignore the relatively small area. Therefore, the extraction of edges and details is a problem that cannot be forgotten. From the classification map, we can see that our method has excellent performance in distinguishing similar classes and preserving edges and details.

Quantitative results obtained by different research methods are shown in Table VII. As can be seen from the table, our method also has the highest classification accuracy in OA, AA, and Kappa coefficients. This also confirms the effectiveness of the method.

3) Houston 2013: The third experiment was conducted on Houston 2013 dataset, and 20 pixels were randomly selected from each class as training samples, accounting for 2% of ground truth. This dataset is more challenging than the first two. The classification maps of all methods are shown in Fig. 11. Just like the previous two datasets, SVM [8] has a serious misclassification of noise labels. The WMRF [26] method with spatial constraint has little effect on improving noise phenomenon.

Three deep methods, i.e., RPNet [51], Spectralformer [53], and WFCG [54], still do not work well in the condition of limited training samples. Besides, the classification effect of RMGE [52] is even worse than that of SVM, which shows that the semi-supervised method may not be very suitable for challenging datasets. In addition, SP [38] and our method are more stable than the above methods, but the classification results of SP method appear over-fitting phenomenon, proving that our method performs better in detail such as small-size targets.

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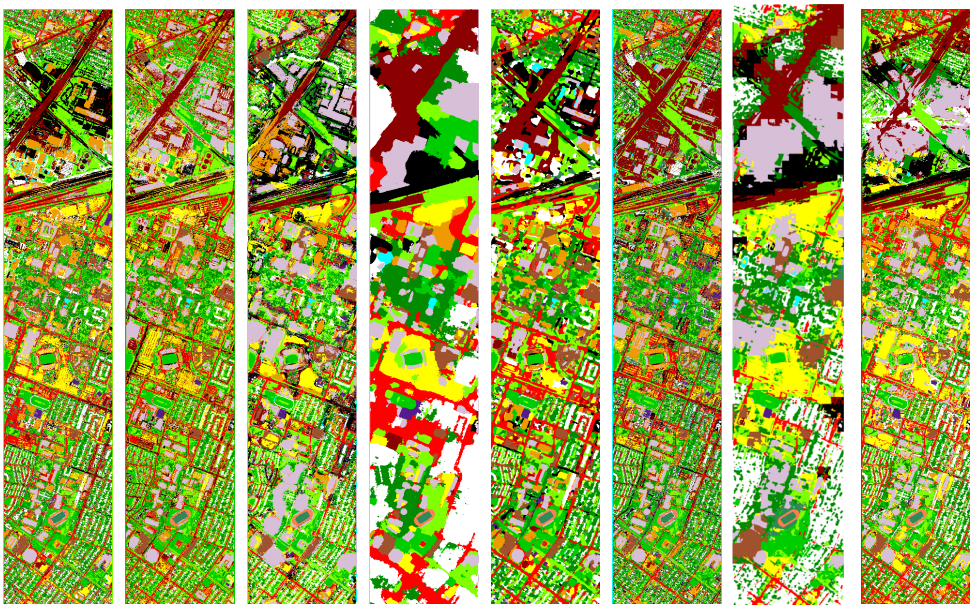

Fig. 11. Classification maps of all studied approaches on the Houston 2013 dataset. (a) SVM [8], (b) WMRF [26], (c) RPNet [51], (d) SP [38], (e) RMGE [52], (f) Spectralformer [53], (g) WFCG [54], and (h) Our method.

TABLE VIII

CLASSIFICATION PERFORMANCE OF ALL STUDIED APPROACHES FOR THE HOUSTON 2013 DATASET, INCLUDING SVM [8], WMRF [26], RPNET [51], SP [38], RMGE [52], SPECTRALFORMER [53], WFCG [54], AND OUR METHOD

Class name	Training set	Testing set	Classification accuracies of different approaches (in %)							
			SVM	WMRF	RPNet	SP	RMGE	Spectralformer	WFCG	Our method
Healthy grass	20	1231	95.42	**98.70**	91.24	86.60	84.97	85.05	96.06	96.66
Stressed grass	20	1234	86.71	96.19	83.57	87.35	70.81	78.61	**97.15**	89.49
Synthetic grass	20	677	99.70	99.85	96.14	**100.00**	99.86	93.06	96.06	**100.00**
Tree	20	1224	**98.37**	92.81	97.16	83.00	63.91	91.75	73.91	97.83
Soil	20	1222	90.20	98.53	94.82	**100.00**	92.51	81.59	90.14	98.86
Water	20	305	94.10	96.39	**98.33**	85.88	98.15	84.59	91.69	92.40
Residential	20	1248	82.58	70.59	91.47	79.94	52.68	62.42	88.04	**91.90**
Commercial	20	1224	88.61	71.81	82.69	**99.18**	44.45	55.72	82.68	87.57
Road	20	1232	66.32	66.96	91.12	85.88	65.10	68.75	**96.24**	89.11
Highway	20	1207	81.10	85.75	88.11	82.88	**93.72**	67.19	81.48	87.43
	20	1215	66.62	65.02	78.45	**97.81**	90.45	64.36	92.31	93.20
Parking lot1	20	1213	65.68	56.97	84.15	**93.94**	66.59	24.57	62.39	86.29
Parking lot2	20	449	32.80	53.45	63.31	80.08	90.19	35.86	**100.00**	77.59
Tennis court	20	408	85.53	**100.00**	91.05	99.76	**100.00**	90.93	**100.00**	90.87
Running track	20	640	97.22	99.06	96.34	**100.00**	97.58	99.69	97.73	98.31
	OA		80.52	82.12	88.13	90.07	76.71	70.57	85.75	**91.96**
	AA		82.07	83.47	88.53	90.82	80.73	72.28	89.73	**91.83**
	Kappa		78.96	80.67	87.17	89.26	74.86	68.24	84.57	**91.31**

Quantitative results obtained by different research methods are shown in Table VIII. It can be seen from the table that the accuracy of the individuals obtained by our method is above 85% in 14 out of 15 kinds of ground objects. Meanwhile, the classification effect of the four similar categories of healthy grass, stressed grass, synthetic grass, and tree is better. Three of the classes had an individual accuracy of around 98% and one class had an individual accuracy of around 90%. Our method also has the highest OA, AA, and Kappa. The above analysis can prove that our method also has a good classification effect for challenging datasets.

H. Computing Cost

In this work, all experiments were carried out with MATLAB R2020B on a laptop computer with Intel(R) Core(TM) I7-10875H CPU processor (2.30 GHz), 16 GB memory, and 64-bit operating system. The running times of all research methods on the three datasets used are shown in Table IX. It can be observed that, in this work, the three experimental datasets in this article are the China dataset, the University of Pavia and Houston 2013 in descending order according to spatial size and spectral dimension. It can also be seen from Table IX that the larger the data scale, the longer the running time, and the two are linearly correlated.

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TABLE IX
COMPUTING TIME OF DIFFERENT APPROACHES

Data sets	Computing time of all studied approaches (in seconds)							
	SVM	WMRF	RPNet	SP	RMGE	Spectralformer	WFCG	Our method
University of Pavia	**3.60**	519.00	7.36	37.12	161.39	343.55	48.41	72.22
The China data set	**6.11**	31.02	7.31	15.05	321.26	713.03	9.63	25.10
Houston 2013	**26.88**	68.46	124.65	158.44	72.24	244.05	85.32	187.14

At the same time, the running time of the proposed method is reasonable. Taking the China dataset as an example, the running time of our method is 25.10 s, which is in the middle of the eight methods compared. In addition, in all the comparison methods, SVM [8] has excellent performance in terms of time, mainly because this method does not need feature extraction or probability optimization of HSI original data like other methods, and directly carries out pixel-level classification of the original image body. This makes the method take less time than the other seven methods, which is understandable because SVM has at least one step less than the other methods.

V. CONCLUSION

In this article, a complementary spectral-spatial method for HSI classification is proposed. It is mainly composed of two stages, i.e., texture smoothing pre-processing feature extraction method based on ACTVSP and post-processing feature extraction method based on edge preserving filtering [39], and then the two probability maps are fused. Our method can smooth the image, effectively remove invalid information, and keep the edge contour information and details from being ignored while extracting features. At the same time, adopting KPCA [49] after ACTVSP can improve the separability between classes, so that the same kind of pixels gather together and different kinds of pixels stay away, which is conducive to improving the classification effect. In addition, our experimental results on three real hyperspectral datasets show that our method has superior performance on both complex and simple datasets. Besides, experiments on ten public datasets show that the generalization and robustness is strong. Moreover, it can also be concluded that when the number of training samples is limited, the classification accuracy of this method is superior to other recently proposed HSI classification methods, and the running time is also in the middle position among the comparison methods.

Although our method has strong robustness for datasets in different situations, but there are still some drawbacks.

1) In some scenarios that require high classification effects, there are six parameters that need to be manually adjusted, which is a little troublesome in actual use and will affect the application to some extent.
2) The running time can be further reduced.
3) The proposed algorithm only considers the spectral adaptive weight without considering the spatial domain adaptive weight.

Based on the drawbacks listed above, the focus of our future research is as follows.

1) How to reduce the parameters that need to be adjusted or how to automatically select the optimal parameters.
2) The ways of reducing algorithm running time and improving execution efficiency by optimizing algorithm.
3) How to combine the adjusted spectral adaptive weight with the spatial domain pixel adaptive weight.

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